# Notes: Liebersohn (JFE, 2024)

## How does competition affect retail banking? Quasi-experimental evidence from bank mergers

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# 1 Big picture

- **Question.** How does local bank competition affect retail banking outcomes after bank mergers, and do antitrust divestiture rules protect consumers?
- **Policy setting.** Bank merger review flags local markets when predicted $\Delta HHI \geq 200$ and predicted post-merger $HHI > 1800$.
- **Quasi-experiment.** The paper compares merger markets around the 1800 HHI cutoff using dynamic difference-in-differences.
- **First stage.** Predicted intervention lowers concentration by about 183 HHI points, or 0.119 log points, and the effect persists for years.
- **Deposits.** CD rates rise by about 0.08–0.11 percentage points, while checking and savings rates barely respond.
- **Mortgages.** Mortgage originations rise by about 11%, and rates fall by about 0.10 percentage points.
- **Small business lending.** Loan counts and volumes do not change detectably.
- **Takeaway.** Local banking competition remains important, but different retail banking products follow different economic models.

# 2 Data and measurement

## 2.1 Banking markets and HHI

The paper uses regulator-defined banking markets from the Federal Reserve Bank of St. Louis CASSIDI team. The HHI is calculated from FDIC Summary of Deposits branch-level deposits, with thrifts receiving half weight under regulatory guidelines.

$$HHI_{m,t} = \sum_{b} s^2_{b,m,t}.$$

**Interpretation.** A higher HHI means a more concentrated local banking market. The key variation comes from predicted post-merger HHI and predicted $\Delta HHI$.

## 2.2 Treatment and control markets

$$Treat_m = 1\{\widehat{\Delta HHI}_m \geq 200,\ \widehat{HHI}^{post}_m > 1800\}.$$

Treated markets are expected to face antitrust intervention and branch divestiture. Control markets are nearby merger markets below the HHI cutoff. The design is intent-to-treat because it uses predicted enforcement.

**Interpretation.** The estimand is the effect of being subject to the antitrust rule, not simply the effect of any observed branch sale.

## 2.3 Outcome data

Data include RateWatch deposit rates, FDIC deposit and branch data, HMDA mortgage origination data, Consumer Credit Panel mortgage rates, and CRA small business loan data.

**Interpretation.** The same regulatory shock is applied to multiple product markets, allowing the paper to test whether competition works differently across CDs, savings/checking, mortgages, and relationship lending.

# 3 Models and empirical specifications

## 3.1 Dynamic difference-in-differences

A simplified version of the estimator is

$$Y_{i,t} = \sum_{c,t} \delta_{c,t} CohortYear_{c,t} + \gamma_i Merger_i + \sum_{c,\tau} \beta_{c,\tau} Post_{c,\tau} \times Treat_i + \varepsilon_{i,t}.$$

**Interpretation.** Merger fixed effects compare treated and control markets in similar merger settings. Cohort-year fixed effects absorb common shocks in a merger cohort-year.

## 3.2 Event-study averages

$$\beta(\tau) = \text{weighted average of } \beta_{c,\tau} \text{ across cohorts at event time } \tau.$$

**Interpretation.** Event studies show timing and persistence; pre-trend tests assess whether treated and control markets were diverging before intervention.

## 3.3 Poisson specifications for counts

$$E[Y_{i,t} \mid X_{i,t}] = \exp(\text{fixed effects} + \beta Post \times Treat + controls).$$

**Interpretation.** Poisson regressions are used for mortgage and small-business loan counts. Coefficients are approximately percentage changes for small coefficients.

# 4 Identification and empirical design

The design exploits the 1800 HHI cutoff. The cutoff is useful because it changes the likelihood of regulatory divestiture without being an economic threshold that should directly change market outcomes. Evidence supporting identification includes no bunching around the threshold, a large increase in branch sales above the threshold, pre-trend tests, and placebo/robustness checks.

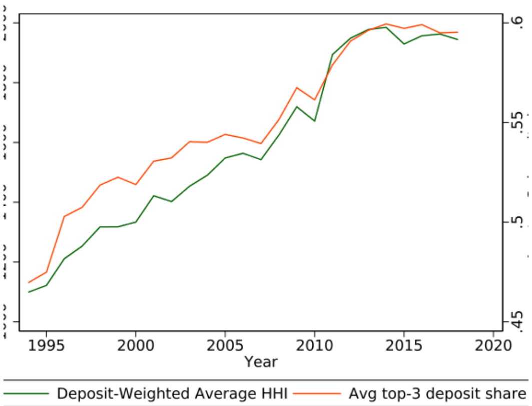
**Interpretation.** The identifying assumption is parallel trends for treated and control markets involved in similar mergers, conditional on merger and cohort-year fixed effects.

# 5 Tables and figures

## 5.1 Figure 1: Average HHI and Top-3 share over time + Table 1: Summary Statistics by Banking Market

**Read:** Figure 1 shows rising local bank concentration. Table 1 shows treated markets are generally smaller and lower-income than non-treated markets, motivating within-merger/cohort comparisons.

**Economic message.** The paper starts from the fact of increasing local concentration, then uses the cutoff to isolate the effect of competition.




1. Average HHI and Top-3 share over time. To create this figure, I calcula
IHI and top-3 bank share by banking market, and take a deposit-weight
ıge of banking markets in each year. Data source: FDIC Summary of I
s database.

(a) Figure 1: Average HHI and Top-3 share over time

**Table 1**

Summary Statistics by Banking Market, Intervention and Non-Intervention mergers. Summary statistics by banking market as of the merger year for mergers with and without predicted intervention. Sample sizes vary depending on data availability. Banking market definitions are provided by the Federal Reserve Bank of St. Louis "CASSIDI" team. Data sources: RateWatch, Community Reinvestment Act, HMDA, FDIC Summary of Deposits, County Business Patterns, IRS Statistics of Income.

| | Mean | Std. Dev. | Min. | Max. | Obs. |
|---|---|---|---|---|---|
| *Panel A: Intervention* | | | | | |
| Payrolls Per Worker | 25.64 | 5.45 | 17.51 | 40.75 | 157 |
| APY, 6-Month CD | 3.41 | 1.49 | 0.07 | 5.26 | 95 |
| APY, 12-Month CD | 3.76 | 1.50 | 0.15 | 5.58 | 96 |
| Business Establishments | 5,553.06 | 12,034.53 | 277.00 | 84,267.00 | 157 |
| Deposits Per Branch | 31,627.32 | 10,863.26 | 9,672.50 | 88,519.50 | 171 |
| Purchase Loans Per Capita (th) | 11.24 | 6.49 | 0.24 | 31.58 | 157 |
| Refi Loans Per Capita (th) | 12.85 | 10.63 | 0.10 | 71.96 | 157 |
| Population | 228,015.82 | 479,624.86 | 6,333.00 | 2,954,274.00 | 157 |
| Income per Capita (th) | 17.86 | 4.26 | 7.79 | 30.95 | 157 |
| Households | 198,734.75 | 424,515.94 | 2,721.00 | 2,635,986.00 | 157 |
| *Panel B: Non-Intervention* | | | | | |
| Payrolls Per Worker (th) | 29.07 | 6.50 | 16.17 | 46.13 | 72 |
| APY, 6-Month CD | 2.92 | 1.60 | 0.10 | 5.04 | 56 |
| APY, 12-Month CD | 3.26 | 1.63 | 0.10 | 5.29 | 56 |
| Business Establishments | 14,573.15 | 26,575.92 | 694.00 | 124,110.00 | 72 |
| Deposits Per Branch | 33,666.47 | 9,784.49 | 15,861.57 | 73,271.88 | 72 |
| Purchase Loans Per Capita (th) | 15.65 | 9.12 | 0.50 | 47.41 | 72 |
| Refi Loans Per Capita (th) | 17.02 | 10.91 | 0.08 | 54.91 | 72 |
| Population | 558,033.44 | 959,185.89 | 24,734.00 | 4,644,327.00 | 72 |
| Income per Capita (th) | 21.28 | 5.23 | 11.58 | 39.07 | 72 |
| Households | 481,795.75 | 824,094.52 | 21,295.00 | 3,908,300.00 | 72 |

(b) Table 1: Summary Statistics by Banking Market

## 5.2 Figure 2: Histogram of post-merger HHIs + Figure 3: Spinoff likelihood by post-merger HHI

**Read:** Figure 2 shows no clear manipulation around the 1800 cutoff. Figure 3 shows branch sales rise above the cutoff.

**Economic message.** The cutoff changes enforcement rather than merger selection.

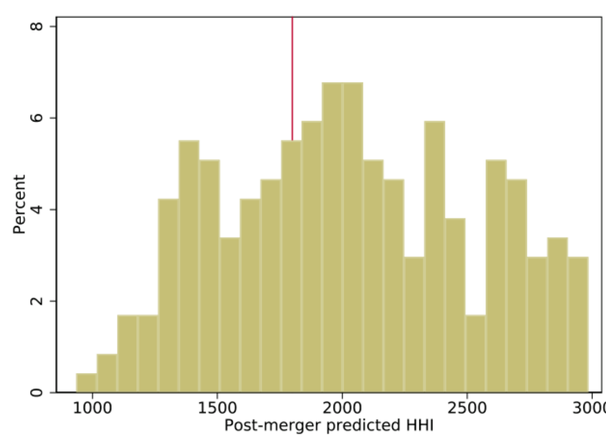
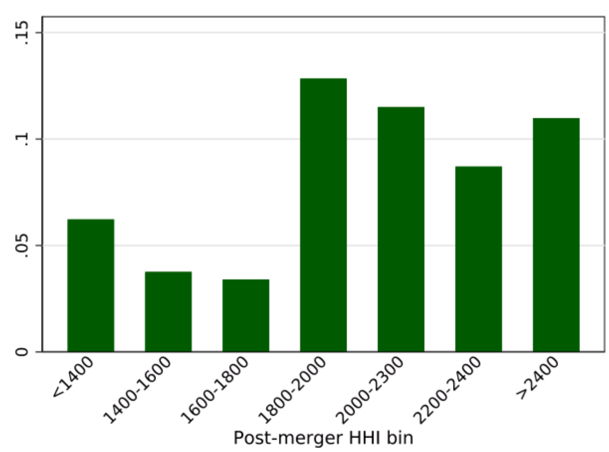



**Fig. 2.** Histogram of post-merger HHIs. Sample includes each merger with a predicted HHI increase of at least 200 points but where the post-merger HHI varies. Mergers with a predicted post-merger HHI above 1800 are the "treated" group in the regression sample. The lack of bunching below the HHI = 1800

(a) Figure 2: Histogram of post-merger HHIs




. **3.** Spinoff likelihood by post-merger HHI. For each merger with a predicted
II increase above 200, I calculate the fraction of branches sold in the year
er the merger. Mergers are binned by their predicted post-merger HHI. Those
h a predicted post-merger HHI above 1800 are the "treated" group in the

(b) Figure 3: Spinoff likelihood by post-merger HHI

## 5.3 Figure 4: Herfindahl Index Event Study + Table 2: Pre-Trend Tests

**Read:** Figure 4 shows lower HHI and log HHI after intervention. Table 2 shows limited evidence of systematic pre-trends.

**Economic message.** Antitrust enforcement has a persistent first-stage effect on concentration, and pre-treatment evidence broadly supports the design.

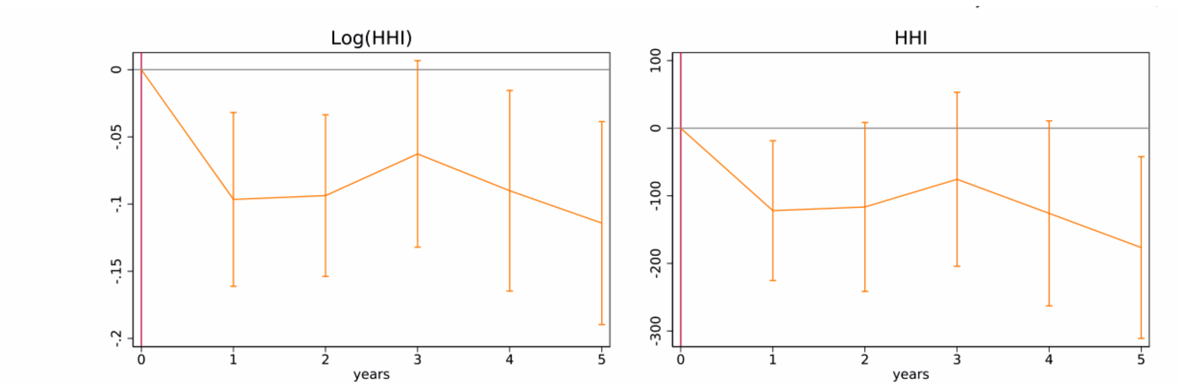



**Fig. 4.** Herfindahl Index Event Study. Event study graph showing the path of the HHI and the natural log of HHI in treated/divestiture and non-tre divestiture banking markets relative to merger year. Event study calculation is described in the text. HHI calculations give thrifts half weight as per re guidelines. Calculations are as described in the text and are calculated from the FDIC Summary of Deposits Database.

Figure 4: Herfindahl Index Event Study

**Table 2**
Pre-Trend Tests. This table tests for statistically significant differences in the main outcome variables between the treated and control markets in the three years prior to the merger, following Wooldridge (2022). The estimates in this table come from specifications in which the independent variables are treatment indicators interacted with pre-merger event-time indicators for the three years prior to the merger year. Columns 1, 2, 9 and 10 are estimated using ordinary least squares, and Columns 3, 4, 5, 6, 7 and 8 are estimated using Poisson regression. The sample is the same as the main estimates but omits post-merger years in treated markets. Standard errors clustered by banking market. See Online Appendix B for more details on the empirical specification and Online Appendix G for further placebo tests.

| | (1) 6m CD | (2) 12m CD | (3) Purchase Ln | (4) Refi Ln | (5) CRA Vol | (6) CRA Lns | (7) Branches | (8) Deposits | (9) Log(HHI) | (10) HHI |
|---|---|---|---|---|---|---|---|---|---|---|
| $I_{t-1} \times TREAT$ | 0.0366 | 0.0424 | 0.102* | 0.0702 | -0.0173 | -0.0330 | 0.106 | 0.0521 | -0.0443 | -106.7 |
| | (0.0710) | (0.0689) | (0.0587) | (0.0505) | (0.0986) | (0.0968) | (0.0917) | (0.0808) | (0.0383) | (70.80) |
| $I_{t-2} \times TREAT$ | -0.0207 | -0.105* | 0.0825* | 0.0691 | 0.0145 | 0.00247 | 0.0728 | 0.0205 | -0.0104 | -84.09 |
| | (0.0623) | (0.0623) | (0.0496) | (0.0550) | (0.0631) | (0.0780) | (0.0805) | (0.0749) | (0.0335) | (66.46) |
| $I_{t-3} \times TREAT$ | 0.0472 | 0.0157 | 0.0401 | 0.0427 | -0.0163 | -0.0287 | 0.0718 | 0.0289 | -0.0149 | -82.08 |
| | (0.0495) | (0.0554) | (0.0380) | (0.0587) | (0.0469) | (0.0564) | (0.0667) | (0.0611) | (0.0286) | (59.48) |
| $N$ | 1459 | 1456 | 2325 | 2315 | 2135 | 2135 | 2585 | 2612 | 2693 | 2693 |
| Cohort-Year FE | X | X | X | X | X | X | X | X | X | X |
| Merger FE | X | X | X | X | X | X | X | X | X | X |
| Chi2-Statistic | . | . | 5.752 | 3.665 | 2.120 | 6.165 | 4.491 | 4.539 | . | . |
| F-Statistic | 0.760 | 2.724 | . | . | . | . | . | . | 1.074 | 0.848 |
| P-Value | 0.519 | 0.0475 | 0.124 | 0.300 | 0.548 | 0.104 | 0.213 | 0.209 | 0.361 | 0.469 |

Table 2: Pre-Trend Tests

## 5.4 Table 3: Deposit Market Herfindahl Index + Table 4: Deposit Rates

**Read:** Table 3 estimates an HHI decline of 182.9 points and a log-HHI decline of 0.119. Table 4 shows CD rates rise, while checking and savings do not respond significantly.

**Economic message.** Competition affects prices most strongly in standardized deposit products.

**Table 3**
Deposit Market Herfindahl Index. Difference-in-differences estimates from an ordinary least squares regression show the effect of predicted antitrust intervention on local bank concentration. These estimates come from a specification which estimates treatment effects for each cohort that mergers take place and for each year in the ten years following the merger. Coefficients shown in the table are the average treatment effect across cohorts and across years, weighted by the number of observations in the sample. See section 5 for more details on the empirical strategy. Columns 1-2 show the effect on the HHI (Herfindahl Index) and the log of HHI as measured using FDIC Summary of Deposits data. Columns 3-4 show HHI estimates limited to markets with above sample median and below sample median population respectively. Standard errors are clustered by banking market.

| | (1)<br>HHI | (2)<br>Log HHI | (3)<br>HHI (small markets) | (4)<br>HHI (large markets) |
|---|---|---|---|---|
| Post X Treat | -182.9*** | -0.119*** | -127.5** | -163.4*** |
| | (58.44) | (0.0336) | (63.02) | (59.07) |
| *N* | 4356 | 4356 | 2103 | 4127 |
| Cohort-Year FE | X | X | X | X |
| Merger FE | X | X | X | X |

(a) Table 3: Deposit Market HHI

**Table 4**
Deposit Rates. Difference-in-differences estimates from an ordinary least squares regression show the effect of predicted antitrust intervention on rates an local bank branches. Regression estimates come from a specification which estimates treatment effects for each cohort that mergers take place and for each year in the ten years following the merger. Coefficients shown in the table are the average treatment effect across cohorts and across years, weighted by the number of observations in the sample. See section 5 for more details on the empirical strategy. Interest rates are a market-level average by product type, calculated using data from Ratewatch. Standard errors are clustered by banking market.

| | (1)<br>6-Month CD | (2)<br>12-Month CD | (3)<br>Interest-Bearing Checking | (4)<br>Savings |
|---|---|---|---|---|
| Post X Treat | 0.0813** | 0.109*** | -0.0635 | -0.0145 |
| | (0.0410) | (0.0378) | (0.0536) | (0.0389) |
| *N* | 2421 | 2417 | 2256 | 2281 |
| Cohort-Year FE | X | X | X | X |
| Merger FE | X | X | X | X |

Savings 12-Month CD

(b) Table 4: Deposit Rates

## 5.5 Figure 5: Deposit rates event study + Table 5: Deposit Quantities

**Read:** Figure 5 shows 12-month CD rates rise while savings rates remain close to zero. Table 5 shows branch counts increase and deposit quantities rise only modestly.

**Economic message.** Consumers benefit in CD markets, but switching costs and account bundling limit the savings/checking response.

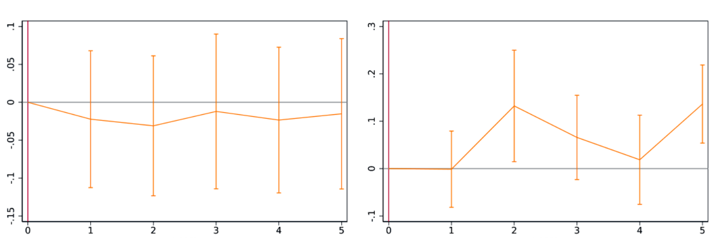



**Fig. 5.** Deposit rates event study. Event study graph showing average deposit rates for 6-month CDs and savings accounts in treated/divestiture bankir
relative to non-treated/divestiture banking markets from the same merger cohort. Event study calculation is described in the text. Interest rate data is
using data from RateWatch.

(a) Figure 5: Deposit rates event study

Deposit Quantities. Difference-in-differences estimates from Poisson regressions show the effect of predicted antitrust intervention on rates an local bank branches. Regression estimates come from a specification which estimates treatment effects for each cohort that mergers take place and for each year in the ten years following the merger. Coefficients shown in the table are the average treatment effect across cohorts and across years, weighted by the number of observations in the sample. Deposit and branch quantities are measured from FDIC Summary of Deposits data. See section 5 for more details on the empirical strategy. Standard errors are clustered by banking market.

| | (1)<br>Deposits | (2)<br>Branches |
|---|---|---|
| Post X Treat | 0.0249 | 0.0573** |
| | (0.0314) | (0.0288) |
| *N* | 3928 | 3846 |
| Cohort-Year FE | X | X |
| Merger FE | X | X |

(b) Table 5: Deposit Quantities

## 5.6 Figure 6: Divestiture and CD Rates and Quantities + Table 6: CD Rates Controlling for Bank Characteristics

**Read:** Figure 6 shows branch counts and deposit quantities over time. Table 6 shows the 12-month CD effect remains around 0.09–0.11 percentage points after controls.

**Economic message.** The CD-rate effect is not explained by observed characteristics of banks acquiring divested branches.

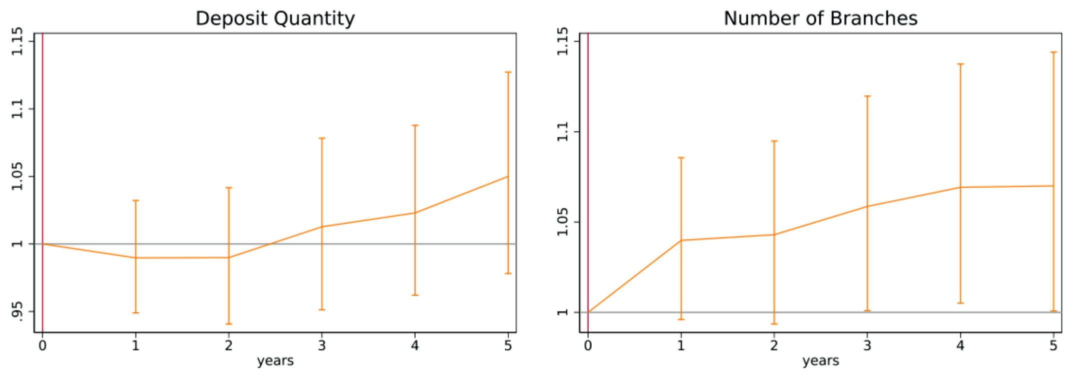



**Fig. 6.** Divestiture and CD Rates and Quantities. Event study graph showing average deposit quantity and branch count in treated/divestiture and non-t divestiture banking markets relative to merger year. Event study calculation is described in the text. Deposits data is calculated from the FDIC Summary o database.

Figure 6: Divestiture and CD Rates and Quantities

**Table 6**

Deposit Rates on Certificates of Deposit, Controlling for Bank Characteristics. Difference-in-differences estimates from an ordinary least squares regression show the effect of predicted antitrust intervention on rates and local bank branches. Estimates come from a specification which estimates treatment effects for each cohort that mergers take place and for each year in the ten years following the merger. Coefficients shown in the table are the average treatment effect across cohorts and across years, weighted by the number of observations in the sample. See section 5 for more details on the empirical strategy. Controls are market averages of bank-level variables for banks which acquire branches through merger or divestiture. Risk controls are loan loss reserves divided by total loans abd non-performing loans divided by total loans. Size control is the log of total bank assets. Efficiency controls are the ratio of noninterest expenses to total assets and the ratio of salary expenses to total assets. Standard errors are clustered by banking market.

| | (1)<br>12-Month CD | (2)<br>12-Month CD | (3)<br>12-Month CD | (4)<br>12-Month CD |
|---|---|---|---|---|
| Post X Treat | $0.104^{***}$ | $0.110^{***}$ | $0.100^{***}$ | $0.0926^{***}$ |
| | (0.0390) | (0.0390) | (0.0370) | (0.0350) |
| $N$ | 2127 | 2136 | 2136 | 2127 |
| Cohort-Year FE | X | X | X | X |
| Merger FE | X | X | X | X |
| Risk Ctrl | X | | | X |
| Size Ctrl | | X | | X |
| Efficiency Ctrl | | | X | X |

Table 6: CD Rates Controlling for Bank Characteristics

## 5.7 Table 7: Mortgage Lending + Table 8: Mortgage Lending, Composition

**Read:** Table 7 shows total mortgage lending rises by about 11%, and mortgage rates fall by about 0.10 percentage points. Table 8 shows stronger effects for low-income and low-DTI borrowers.

**Economic message.** Mortgage markets respond like transactional credit markets: more competition raises quantity and lowers rates.

**Table 7**

Mortgage Lending. Difference-in-differences estimates from a Poisson regression show the effect of predicted antitrust intervention on total mortgage lending volumes and rates. Estimates come from a specification which estimates treatment effects for each cohort that mergers take place and for each year in the ten years following the merger. Coefficients shown in the table are the average treatment effect across cohorts and across years, weighted by the number of observations in the sample. See section 5 for more details on the empirical strategy. Loan quantities measured using HMDA data. In Column 1, the dependent variable is the total number of loans. In Columns 2 and 3, it is the refinancing and purchase loans respectively. In Columns 4 and 5, it is total loans from nonbank and local lenders respectively. In Column 6, it is the residualized interest rate calculated from credit registry data. Standard errors are clustered by banking market.

| | (1) Total | (2) Refinance | (3) Purchase | (4) Non-Bank | (5) Local | (6) Rates |
|---|---|---|---|---|---|---|
| Post X Treat | 0.106*** | 0.111** | 0.0860** | 0.119*** | 0.109** | -.0997** |
| | (0.0364) | (0.0442) | (0.0353) | (0.0427) | (0.0495) | (0.0479) |
| *N* | 3965 | 3963 | 3972 | 3984 | 3905 | 1399 |
| Cohort-Year FE | X | X | X | X | X | X |
| Merger FE | X | X | X | X | X | X |

(a) Table 7: Mortgage Lending

**Table 8**

Mortgage Lending, Composition. Difference-in-differences estimates from a Poisson regression show the effect of predicted antitrust intervention on mortgage lending volumes by mortgage and customer type. Estimates come from a specification which estimates treatment effects for each cohort that mergers take place and for each year in the ten years following the merger. Loan quantities measured using HMDA data. Column 1 estimates the effects of predicted antitrust enforcement on the number of loan applications using a Poisson regression. Columns 2-5 estimate the effect of predicted enforcement on the number of loans for different samples of loans and individuals, as noted by column. "Post" is an indicator equal to 1 following mergers, "Treat" is an indicator equal to 1 in markets which I calculate should have antitrust intervention, and "Period" is an indicator equal to 1 in 2010 or later. "Low Inc" refers to individuals who report an income of below $100,000 and "High Inc" have an income above this. "Low DTI" and "High DTI" refer to loans above and below the median DTI. Standard errors clustered by banking market.

| | (1) Applications | (2) Low Inc | (3) High Inc | (4) Low DTI | (5) High DTI |
|---|---|---|---|---|---|
| Post X Treat | 0.113*** | 0.120*** | 0.0887** | 0.117*** | 0.0751 |
| | (0.0412) | (0.0440) | (0.0435) | (0.0403) | (0.0520) |
| *N* | 3963 | 3961 | 3939 | 3967 | 3962 |
| Cohort-Year FE | X | X | X | X | X |
| Merger FE | X | X | X | X | X |

(b) Table 8: Mortgage Lending, Composition

## 5.8 Figure 7: Number of Refinancing Mortgages and Purchase Mortgages

**Read:** Refinancing and purchase loans both rise, but refinancing responds more slowly.

**Economic message.** Dynamic event studies matter because refinancing depends on interest-rate timing.

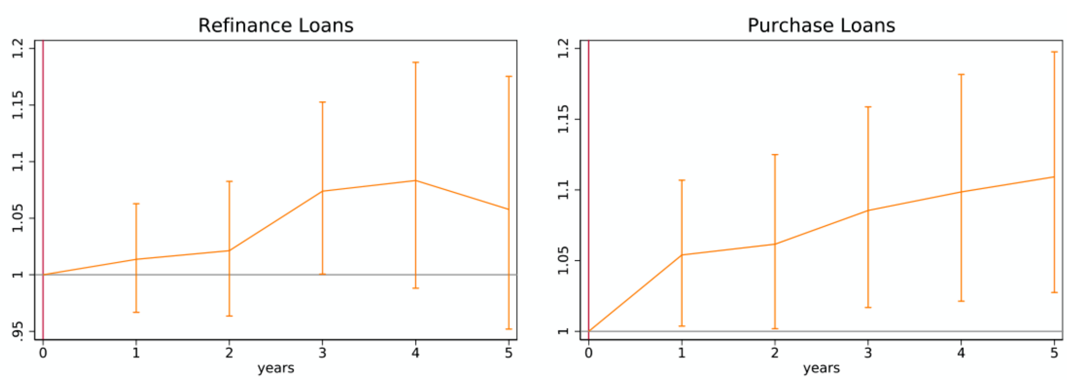



**Fig. 7.** Number of Refinancing Mortgages (left) and Purchase Mortgages (right). Both are event study graphs showing average mortgage origination l treated/divestiture and non-treated/no-divestiture banking markets relative to merger year. Event study is estimated using a Poisson regression as descri text. Mortgage data is calculated using raw mortgage data from HMDA.

Figure 7: Number of Refinancing Mortgages and Purchase Mortgages

### 5.9 Table 9: Small Business Lending + Figure 8: Small Business Lending

**Read:** Table 9 and Figure 8 show no detectable effect on small business lending counts or volume.

**Economic message.** Relationship lending insulates small-business credit from local concentration changes.

**Table 9**
Small Business Lending. Difference-in-differences estimates from a Poisson regression show the effect of predicted antitrust intervention on rates an local bank branches. Estimates come from a specification which estimates treatment effects for each cohort that mergers take place and for each year in the ten years following the merger. Coefficients shown in the table are the average treatment effect across cohorts and across years, weighted by the number of observations in the sample. See section 5 for more details on the empirical strategy. Small business lending is measured using Community Reinvestment Act (CRA) data. In Columns 1-2 the unit of observation is a bank marking, and in Columns 3-4 it is a ZIP code. The dependent variable in columns 1 and 3 is the number of loans and in columns 2 and 4 it is the loan dollar amount. Standard errors are clustered by banking market.

| | (1) Number of Loans | (2) Loan Volume | (3) Number of Loans | (4) Loan Volume |
|---|---|---|---|---|
| Post X Treat | -0.00776 | -0.00497 | 0.0105 | 0.00399 |
| | (0.0423) | (0.0417) | (0.0401) | (0.0444) |
| *N* | 3881 | 3881 | 16288 | 16288 |
| Cohort-Year FE | X | X | X | X |
| Merger FE | X | X | X | X |
| Geography | Market | Market | ZIP | ZIP |

(a) Table 9: Small Business Lending

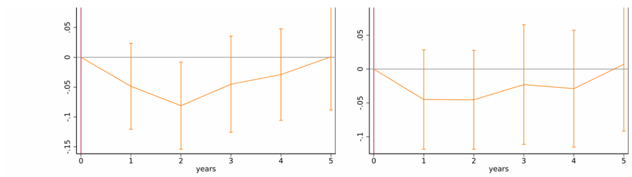



**Fig. 8.** Small Business Lending. Event study graph showing the number of small business loans to businesses in treated/divestiture and non-treated/no-banking markets, relative to merger year. Event study point estimates come from a Poisson regression which estimates treatment effects for each cohort th take place and for each year in the ten years following the merger. Coefficients shown in the table are the average treatment effect across cohorts w year relative to merger year, weighted by the number of observations in the sample. See section 5 for more details. Data is from the Community Reinve: database.

tion is beyond the scope of this paper, but is a promising area for future research.

These results support a view of banking in which differ els apply to different markets. The "structure-conduct-perf

(b) Figure 8: Small Business Lending

## 6 Short summary

- Antitrust intervention lowers local banking concentration by about 183 HHI points.
- CD rates rise substantially, but checking and savings rates do not.
- Mortgage originations rise and mortgage rates fall.
- Small business lending is unaffected, consistent with relationship-lending frictions.
- Local banking competition remains important even with online and nonbank lenders.

## 7 Conclusion

### 7.1 Core conclusion

Bank antitrust intervention can reduce local concentration and improve consumer outcomes, but the effects differ across banking products.

### 7.2 Economic intuition

CDs and mortgages are closer to transactional products; small business loans are relationship-intensive. Competition therefore changes prices and quantities in some markets but not all.

### 7.3 Policy implication

Antitrust divestitures are useful for consumer retail products, but policies targeting relationship lending may need tools beyond concentration-based merger review.

### 7.4 Boundary conditions

The results are local to merger markets near the HHI threshold and should not automatically be extrapolated to all banking markets.

### 7.5 Takeaway

The paper supports a product-specific view of bank competition: local competition is still important, but the relevant economic mechanism depends on the banking product.